

ENV 5128

Smart Air Quality Monitoring & Air Pollution Control

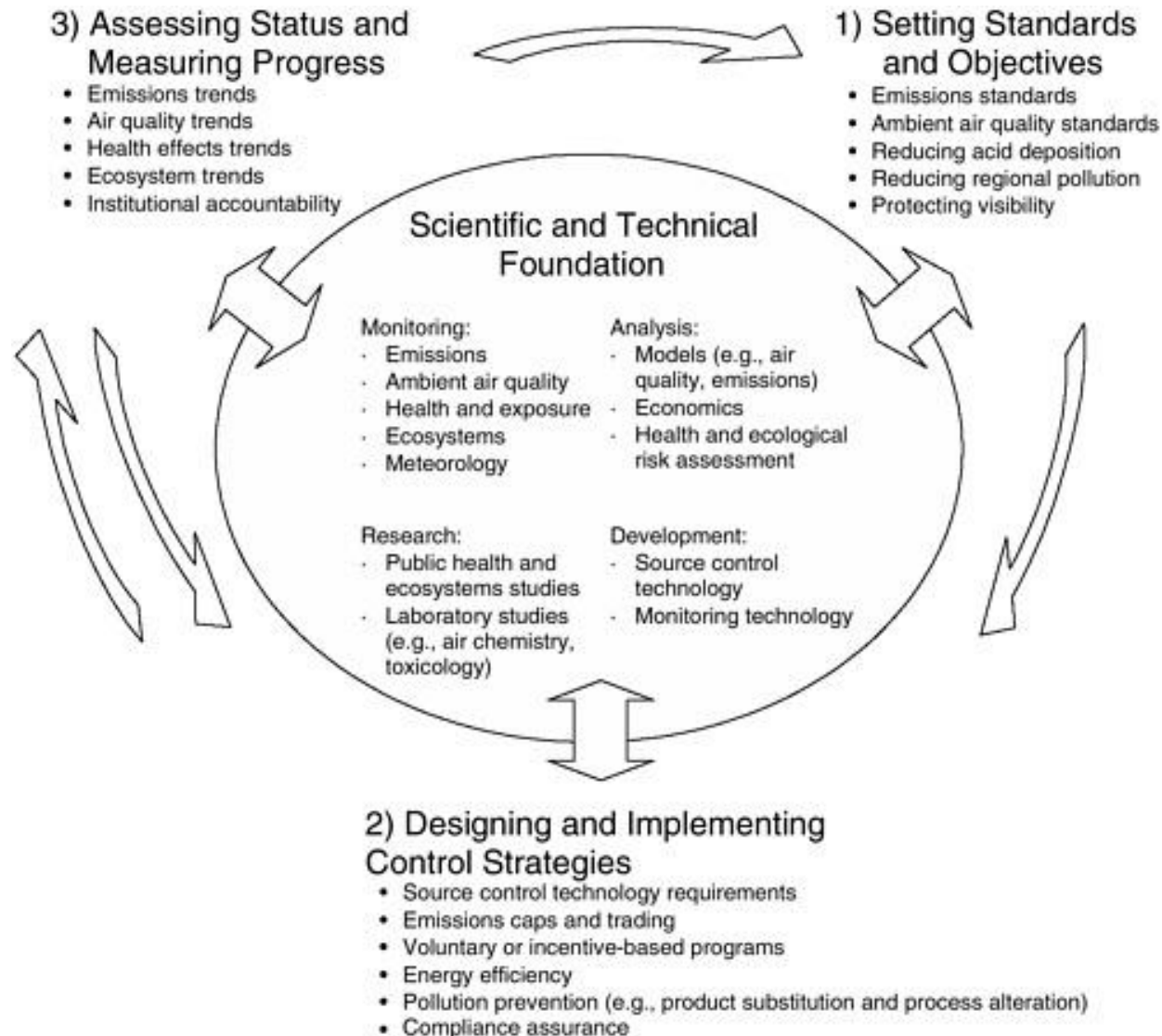
Today's Topics

- Introduction to air pollution
- Expectation
 - Understand how ambient air pollution is managed in the US

Ambient air quality management

Clean Air Act (CAA)

- First established in 1963, amended multiple times
- Primary & secondary standards:
 - Primary: for the protection of human health
 - Secondary: for the protection of public welfare
- National Ambient Air Quality Standards (NAAQS)
 - Regulate six Criteria Air Pollutants (CAPs)
- National Emissions Standards for Hazardous Air Pollutants (NESHAPS)
 - Regulate Hazardous Air Pollutants (HAPs)



The iterative nature of air quality management

<https://www.nap.edu/catalog/10728/air-quality-management-in-the-united-states>

Introduction

- Air quality is a public good
 - Shared resource, value distributed over population
- Air pollutant emissions result from many activities that are beneficial to quality of life
 - It is not possible or desirable to remove all emissions
- Costs and benefits are often unequally distributed

Introduction

- External cost are often external to the market
 - Environmental damage, health impact etc.
 - Difficult to calculate accurately
 - Difficult to attribute in a meaningful way to individual purchaser
- The Tragedy of the Commons
- Regulations are “unproductive resources”
- Much work on trying to figure out how to provide incentives that work with/not against market

Non-Regulatory Legal Remedies

- Tort Law
 - Individuals file civil suit against source to recover damage or to abate the problem
 - Nuisance and trespass principles used to air pollution historically
- Nuisance
 - Act resulting in “unreasonable” interference with use and enjoyment of one’s property
- Trespass
 - Physical violation of land owners right to exclusive possession
 - Used for smoke historically

Regulatory Air Quality Management Strategies

- All regulatory and policy measures implemented in an attempt to control concentrations and effects of air pollutants
- Usually refers to control by some political entity
 - HC EPC, FL DEP, US EPA, US OSHA, US FDA etc...
- Types of management strategies:
 - Air quality standards, emissions standards, economics-based approaches

Ambient Air Quality Standards

- Levels of pollutant concentrations in air that are allowable/acceptable
 - Generally set at a level considered not harmful
 - Based on principle of threshold levels
 - Not necessarily appropriate for all pollutants
 - Provide “margin of safety” for sensitive populations
 - Lower than occupational values set for healthy adults

Ambient Air Quality Standards

- Issues with ambient air quality standards
 - *Require air quality monitoring to determine levels*
 - *Local exposures not adequately represented*
 - Doesn't deal with how to achieve these levels
 - Control and management practices must be implemented to achieve standards
 - This is generally fairly complicated undertaking

Emissions Standards

- Levels of acceptable emissions
- Based often on technological achievability and economic costs
- Simpler to manage
- Problematic because not necessarily related to actual air quality

Economics-Based Approaches

- Provide a system that provides incentives for decreasing pollution
- Include cost-benefit analysis and market-based policies

Cost-Benefit Analyses

- Attempt to determine the appropriate level of pollution
- Industry groups are very interested in having cost-benefit analyses performed
- Problems
 - Difficult to quantify all external costs
 - Loss of sense of community, changes in ecosystems etc..
 - Very difficult to deal with equity

Market-based policies

- Market-based policies involves the bubble concept, emission trading and allowance
- Bubble concept
 - Existing plants or groups of plants are considered one entity from emissions point of view
 - Facilities have flexibility to increase emissions of one process if overall emissions are the same or lower.
 - Flexibility can lead to most cost effective reduction strategies without unreasonably hampering process innovation

Market-based policies

- Facility is given an “emission allowance”
- Excess emission reduction → Emissions reduction credits
 - The credits may be sold or traded to other facilities
- Apply economic value to emissions reductions
- Effective method

Pollution charges

- Less used method
- Charges by amount of emissions
- Difficult to set appropriate charges
- Must be enough to stimulate reductions without being unreasonably burdensome

Brief overview of Clean Air Act

Titles of Clean Air Act (90)

Title I - Air Pollution Prevention and Control

Part A NAAQS, SIP, NSPS, NESHAPS

Part B (replaced by Title VI)

Part C Prevention of Significant Deterioration (PSD)

Part D Non-Attainment area requirements

Title II - Mobile Sources

Part A Motor-vehicle emission and fuel standard

Part B Aircraft emission standard

Part C Clean fuel vehicles

Title III - General (Administrative) (originally Air Toxics)

Title IV - Noise and Acid Deposition Control (IV-A)

Title V - Operating Permits

Title VI - Stratospheric Ozone Protection

Others: Enforcement, Miscellaneous etc

(Codified under 40 CFR Part 50-99)

National Ambient Air Quality Standards (NAAQS)

- 1970 CAA authorized EPA to establish NAAQS.
- Primary & secondary standards:
 - Primary: for the protection of human health
 - Secondary: for the protection of public welfare
- Six Criteria Air Pollutants (CAPs):
 - O₃, CO, NO₂, SO₂, Pb, PM (PM_{2.5} & PM₁₀)
- Air Quality Criteria documents present scientific evidence linking pollutant so health and welfare effects (for establishing the appropriate standards)

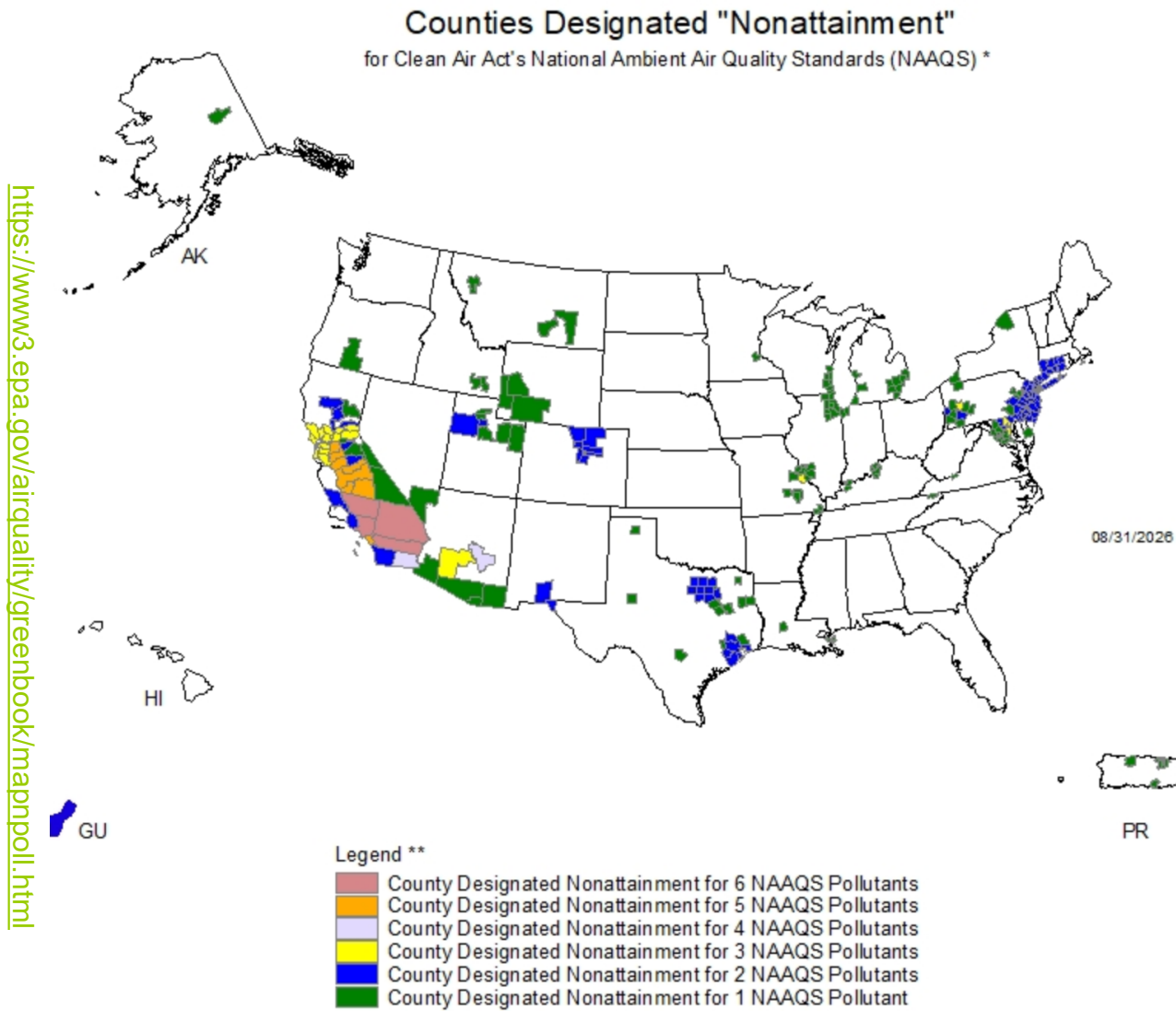
State Implementation Plans (SIPs)

- States required to develop and submit plans for the implementation, enforcement and maintenance of NAAQS
- Plans include
 - Legal authority for regulation and compliance
 - Emissions limitations and control requirements
 - Permitting
 - Monitoring, modeling,
 - Data collection and analysis
 - Episode Control Plans:
 - Provisions for an air pollution episode or emergency

Non Attainment Areas

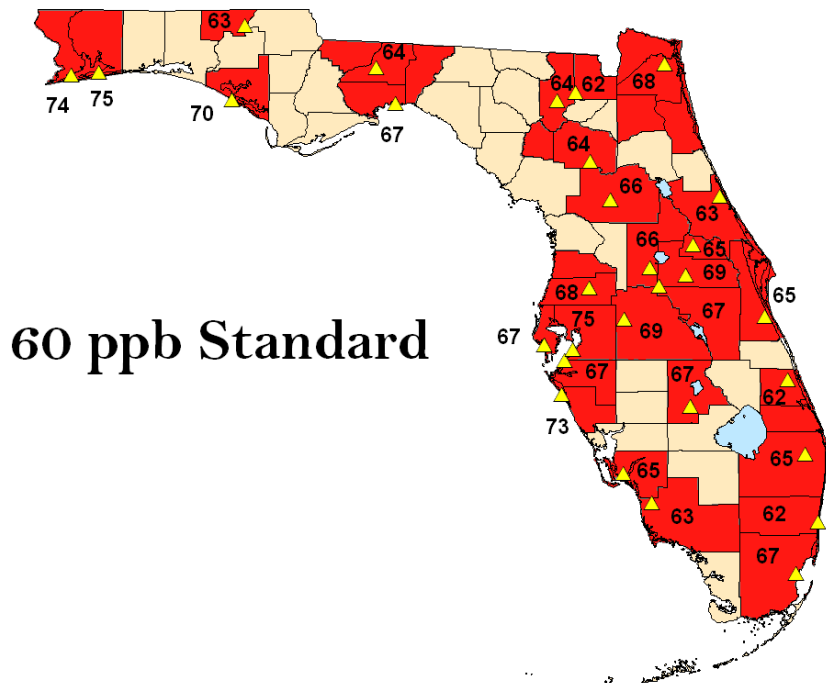
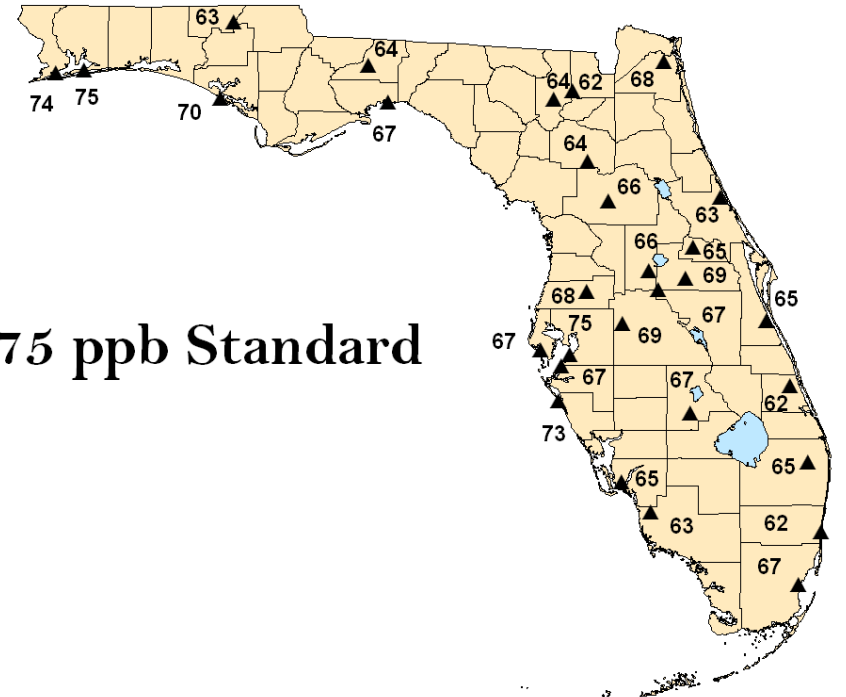
- Areas not in attainment of NAAQS are designated as Non-attainment Areas
- Significant because does not allow new sources (theoretically), since dangerous to health already
- Concerns with limits on economic growth in non-attainment areas have led to
 - Emissions offset policy (no net increase from a facility)
 - Alternative of Lowest Achievable Emission Rate for new sources
 - Postponement of deadlines for attainment

Non Attainment Areas



Non Attainment Areas

USEPA was planning to reduce
O₃ std to 0.06-0.07 ppm
Florida Nonattainment under
the new O₃ Standard



Prevention of Significant Deterioration (PSD)

- To protect and enhance clean air areas
- Divides areas in attainment (for specific pollutants) into:
 - Class III: allow deterioration to NAAQS secondary standard
 - Class II: allow moderate deterioration (but not above NAAQS)
 - Class I areas: international park, national wilderness areas, national monuments, national parks (> about 5000 acres in size)
 - 156 Class I areas
 - Regional Haze Rules to improve visibility by 2008
 - Include BACT for new sources and BART (R=retrofit) technologies for existing sources

http://www.epa.gov/ttn/oarpg/t1/fr_notices/classimp.gi



Air Toxics or Hazardous Air Pollutants (HAPs)

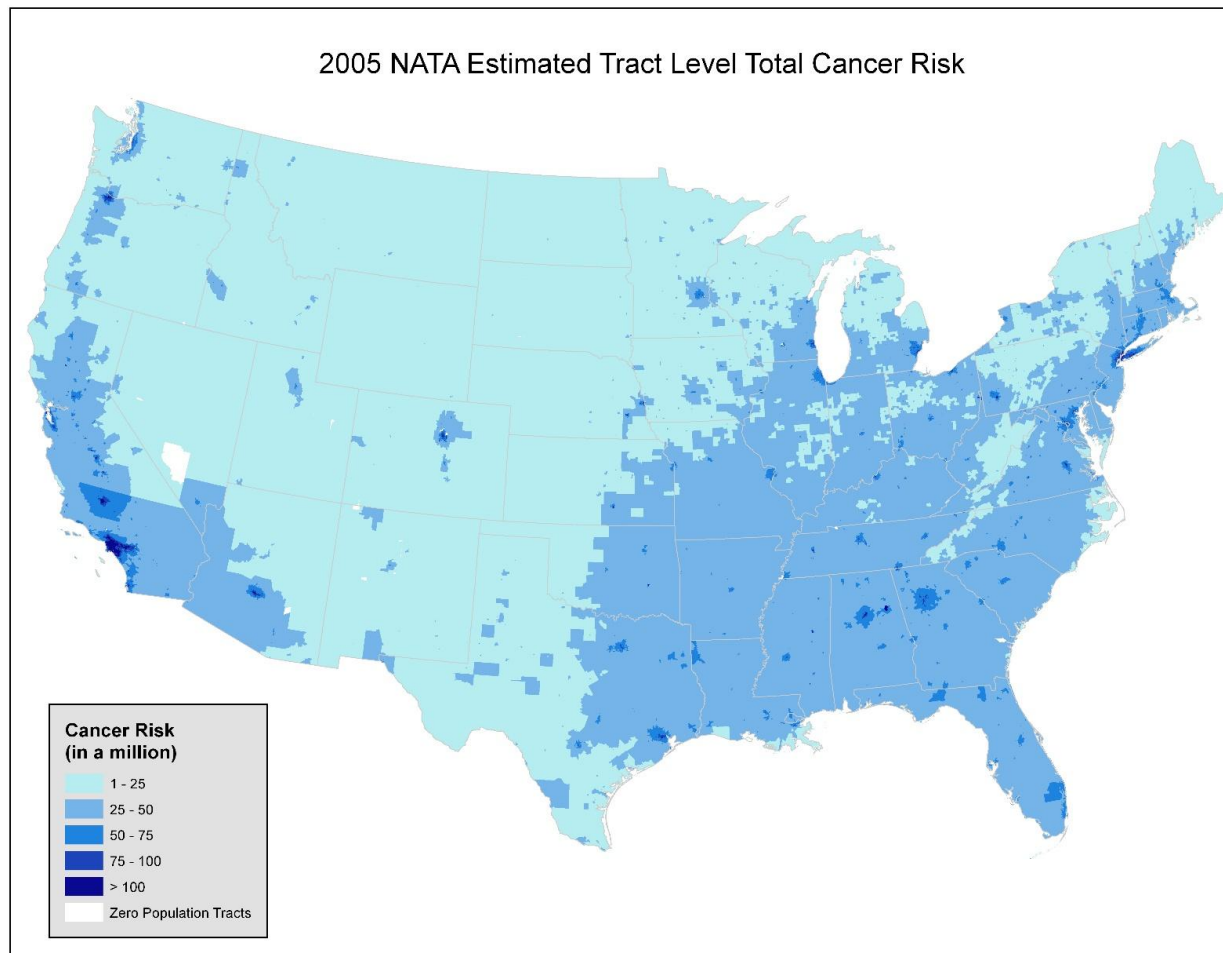
- Pollutants known or suspected to cause serious health or environmental effects (e.g. carcinogens, mutagens)
 - HAPs (also called Air Toxics)
- 187 pollutants designated
 - Organics, chlorinated organics, acid gases, sulfur compounds, metals
- EPA must identify source categories and promulgate emissions standards
 - Major source: >10 tons/yr of single HAP or 25 tons/yr total

Air Toxics or Hazardous Air Pollutants (HAPs)

- National Emissions Standards for Hazardous Air Pollutants (NESHAPS)
 - Technology based emissions standards for specific stationary source categories and pollutants
 - Maximum Achievable Control Technology (MACT)
 - Both process changes and control technology
- Establish program for prevention of accidental releases
- National Air Toxics Assessments (NATA)
 - EPA's ongoing comprehensive evaluation of air toxics
- National Toxics Inventory
 - now part of National Emissions Inventory

National Air Toxics Assessments (NATA)

<http://transportpolicy2013.blogspot.com/2013/06/stop-take-step-for-new-healthy-air-in.html>



**See latest 2020 Air Toxics Screening Assessment
(<https://www.epa.gov/AirToxScreen/airtoxscreen-mapping-tool>)**

New source rules

- New Source Performance Standards (NSPS)
 - Technology-based emissions standards specific to source categories
 - E.g. fossil fuel-fired steam generators, incinerators, iron and steel plants
 - For new or significantly modified sources
 - Criteria pollutants, HAPs, others ...
 - Apply whether a source is in a clean or a dirty region
- New Source Review (NSR)
 - Permitting program for new or significantly modified sources
 - Must have permit before construction begins (1977 CAA)

Operating Permits

- All existing major stationary sources must have a permit
- Recording in one document all air pollutions requirements
- Permits contain
 - enforceable emission limits
 - monitoring and reporting requirements
 - compliance plan with an enforceable schedule
- Burden of proof for compliance is on the emission source
- Permit fees to pay for regulatory programs
- Review and renewal every 5 years

[FLDEP map:](https://floridadep.gov/air/air-business-planning/content/map-active-facilities-current-air-permits)

<https://floridadep.gov/air/air-business-planning/content/map-active-facilities-current-air-permits>

Acid Deposition

- Control of power plant emissions
- Reductions of sulfur dioxide (SO₂) and nitrogen oxides (NO_x) emissions
- Market based system
 - Emissions allowances for SO₂
 - Trading and banking
 - Sources have compliance plans
- Acid rain permits

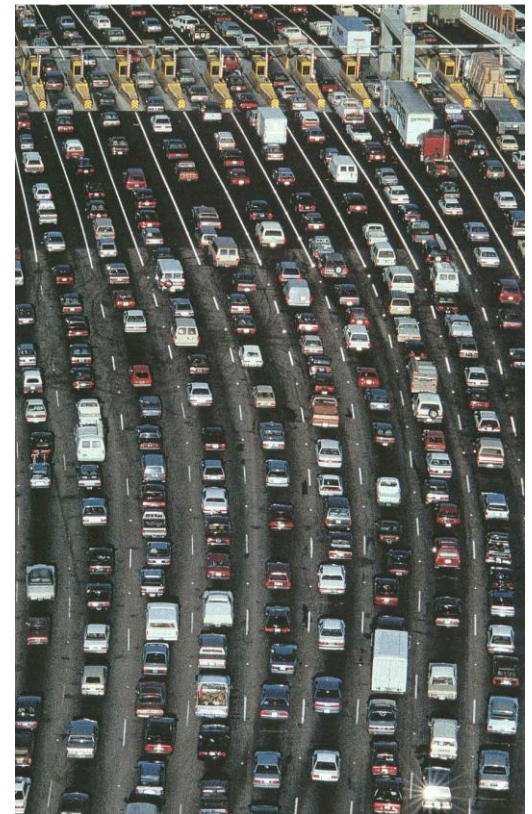


Stratospheric Ozone

- Phase-out schedule
 - Class I - starting in 2000
 - CFCs, Br-halons, carbon tetrachloride, methyl chloroform
 - Class II - starting in 2015
 - HCFCs
- Monitoring, reporting and labeling requirements
- National recycling program
- Development of safe alternatives

Mobile Source Controls

- Motor vehicle emissions and fuel standards
 - Exhaust gas standards
 - Fuel additives and cleaner fuels
 - Inspection and maintenance
 - Tampering prohibitions
 - Onboard vapor recovery systems
 - Onboard diagnostic systems
- Aircraft emissions
- Clean fuel vehicles and fleets



Enforcement

- CAA contains a broad array of authorities to make the law more readily enforceable
- Fines up to 1 million, prison term up to 15 years
- Field citations up to \$5,000
- Enhanced civil judicial penalties
- Criminal penalties upgraded to felonies
- Sources must certify compliance
- Citizens can seek penalties against violators

International Conventions for Air Pollution Control

Agreement	Location (nickname)	Date
Long-range Transboundary Air Pollution	Geneva Convention	Nov 13, 1979
Substances that deplete to ozone layer	Montreal Protocol	Sept 16, 1987
Greenhouse gas emissions reductions	Kyoto Protocol	Dec. 11, 1997
	Paris Agreement	Apr. 22, 2006
Ship emissions	MARPOL Protocol	Sept, 1997
Persistent Organic Pollutants	Stockholm Convention	May 13, 2001

Some important international conventions for air pollution control